

# L23 — Stack Operations: PUSH & POP; Parenthesis Matching

Unit: 2 | Chapter: Stacks & Recursion | BT Level: BT5 | CO: CO2, CO3

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## Learning Objectives

- Trace PUSH and POP operations step by step
  - Implement the parenthesis matching algorithm using a stack
  - Extend matching to multiple bracket types
  - Apply stack-based validation to real programming scenarios
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## 1. Detailed PUSH and POP Operations

### 1.1 PUSH (Add to Top)

```
Algorithm PUSH(stack, item):  
    if stack.top == capacity - 1:  
        raise Overflow  
    stack.top ← stack.top + 1  
    stack[stack.top] ← item
```

**Trace:** Push 5, 10, 15 into empty stack (capacity=5):

#### Operation top Stack State

|          |    |             |
|----------|----|-------------|
| Initial  | -1 | []          |
| push(5)  | 0  | [5]         |
| push(10) | 1  | [5, 10]     |
| push(15) | 2  | [5, 10, 15] |

### 1.2 POP (Remove from Top)

```
Algorithm POP(stack):  
    if stack.top == -1:  
        raise Underflow  
    item ← stack[stack.top]  
    stack.top ← stack.top - 1  
    return item
```

**Trace continued:**

#### Operation top Stack State Returned

|       |   |         |    |
|-------|---|---------|----|
| pop() | 1 | [5, 10] | 15 |
| pop() | 0 | [5]     | 10 |

|       |    |    |            |
|-------|----|----|------------|
| pop() | -1 | [] | 5          |
| pop() | —  | —  | Underflow! |

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## 2. Parenthesis / Bracket Matching

**Problem:** Given a string containing brackets `()`, `[]`, `{}`, determine if the brackets are properly matched and nested.

**Valid:** `()`, `() [] {}`, `{ [ () ] }`, `(( ( )))`

**Invalid:** `[]`, `( [ ] )`, `{, , ) )`

### 2.1 Algorithm

```
Algorithm is_balanced(s):
    stack = empty stack
    for each char c in s:
        if c is an opening bracket: push(c)
        elif c is a closing bracket:
            if stack is empty: return False    (no matching opener)
            top = pop()
            if top and c don't match: return False
    return stack is empty    (all openers matched)
```

### 2.2 Implementation

```
def is_balanced(s):
    stack = []
    matching = {')': '(', ']': '[', '}': '{'}

    for char in s:
        if char in '([{':
            stack.append(char)
        elif char in ')]}':
            if not stack:
                return False    # Nothing to match
            if stack[-1] != matching[char]:
                return False    # Mismatched brackets
            stack.pop()

    return len(stack) == 0    # True if all openers were matched
```

### 2.3 Detailed Trace — `{ [ ( ) ] }`

| Step | Char | Action              | Stack      |
|------|------|---------------------|------------|
| 1    | {    | Push                | [{]        |
| 2    | [    | Push                | [{, [ ]    |
| 3    | (    | Push                | [{, [, ( ] |
| 4    | )    | Pop ( — matches ) ✓ | [{, [ ]    |

```

5   ]   Pop [ — matches ] ✓ [{}]
6   }   Pop { — matches } ✓ []

```

Stack is empty → **Balanced ✓**

## 2.4 Trace — ([)] (Invalid)

| Step | Char | Action                                    | Stack    |
|------|------|-------------------------------------------|----------|
| 1    | (    | Push                                      | [ ( ]    |
| 2    | [    | Push                                      | [ (, [ ] |
| 3    | )    | Top is [, expected ( → <b>mismatch!</b> — |          |

→ **Not balanced ✗**

## 3. Extended Applications

### 3.1 HTML/XML Tag Matching

```

def check_html_tags(html):
    """Check that HTML opening and closing tags are properly nested."""
    import re
    stack = []
    # Find all tags: <div>, </div>, <p>, etc.
    tags = re.findall(r'</?[a-zA-Z]+>', html)

    for tag in tags:
        if not tag.startswith('</'):
            # Opening tag — extract name and push
            tag_name = tag[1:-1]
            stack.append(tag_name)
        else:
            # Closing tag
            tag_name = tag[2:-1]
            if not stack or stack[-1] != tag_name:
                return False
            stack.pop()

    return len(stack) == 0

print(check_html_tags("<div><p></p></div>")) # True
print(check_html_tags("<div><p></div></p>")) # False

```

### 3.2 Minimum Bracket Additions to Make Valid

```

def min_additions_to_balance(s):
    """
    Returns minimum number of brackets to add/remove to balance s.
    """

```

```

open_count = 0    # Unmatched '('
close_count = 0   # Unmatched ')'

for c in s:
    if c == '(':
        open_count += 1
    elif c == ')':
        if open_count > 0:
            open_count -= 1    # Matched
        else:
            close_count += 1  # Unmatched closing

return open_count + close_count

```

### 3.3 Score of Parentheses (LeetCode #856)

```

def score_of_parentheses(s):
    """
    () = 1 point, (A) = 2*score(A), AB = score(A) + score(B)
    """
    stack = [0]    # Start with 0 (represents outer scope)
    for c in s:
        if c == '(':
            stack.append(0)
        else:    # c == ')'
            v = stack.pop()
            stack[-1] += max(2 * v, 1)
    return stack[0]

print(score_of_parentheses("()"))    # 1
print(score_of_parentheses("(()))")) # 2
print(score_of_parentheses("(())()")) # 2
print(score_of_parentheses("((()))")) # 4

```

## 4. Complexity Analysis

| Algorithm            | Time   | Space             |
|----------------------|--------|-------------------|
| Is Balanced          | $O(n)$ | $O(n)$ worst case |
| Min Additions        | $O(n)$ | $O(1)$            |
| Score of Parentheses | $O(n)$ | $O(n)$            |

Each character is pushed/popped at most once  $\rightarrow O(n)$  total operations.

## 5. Stack Size Analysis

The maximum stack size during parenthesis matching equals the **maximum nesting depth**:

- `((((( ))) ) )` — depth 5  $\rightarrow$  max stack size 5

- `() () () ()` — depth 1  $\rightarrow$  max stack size 1

For a string of length  $n$ , worst case stack size is  $n/2$  (all opening brackets first)  $\rightarrow O(n)$  space.

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## Summary

- **PUSH:**  $O(1)$  — increment top and place element.
  - **POP:**  $O(1)$  — retrieve element and decrement top.
  - Parenthesis matching: scan left to right, push openers, pop on closer and verify match.
  - Valid if and only if: every closer matches its opener AND stack is empty at end.
  - Time:  $O(n)$ , Space:  $O(n)$  for the stack.
  - Extends naturally to HTML validation, expression parsing, etc.
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## References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.1 (MIT Press, 2022)
- LeetCode #20 (Valid Parentheses), #856 (Score of Parentheses)